



Short-term traffic prediction based on time series decomposition

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ARTICLE INFO

Article history:

Received 26 May 2021

Received in revised form 9 September 2021

Available online 15 September 2021

Keywords:

Short-term traffic flow prediction

Time series decomposition

Periodic pattern

Multi-step prediction

ABSTRACT

Traffic flow decomposition is an alternative method to explore the composition of traffic flow and improve prediction accuracy. However, most of them suffer from the inability to fully utilize the character of traffic data. This paper presents a novel framework for traffic flow decomposition and modeling named Time Series Decomposition (TSD). The traffic flow is adaptively decomposed into periodic component, residual component and volatile component which are modeled respectively. Empirical Mode Decomposition (EMD) is applied to extract the intrinsic mode functions (IMFs) of traffic flow, the periodic patterns are intuitively presented via Hilbert transform in terms of frequencies. Then the periodic component can be described as a Fourier series based on obtained frequencies. Meanwhile, the residual component is presented by IMF with the lowest frequency. The remaining part is the volatile component modeled by supervised learning. The proposed hybrid model is evaluated on the real-world dataset and compared with classical baseline models. The results demonstrate that TSD can unearth the underlying periodic patterns and provide an explicable composition of the traffic flow. Furthermore, the volatile component ensures the accuracy of single-step prediction while periodic and residual components show promising abilities in improving the multi-step prediction accuracy of short-term traffic flow.

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1. Introduction

Short-term traffic flow prediction plays a crucial role in advanced Intelligent Transportation System (ITS) as the ITS depends on short-term (within 15 min) information to describe the evolution of traffic flow over time and must therefore obtain accurate and timely traffic data [1]. In contrast to medium-term and long-term prediction, accurate short-term prediction is still a challenging topic because of the acute volatility induced by ultrashort interval. Short-term prediction of traffic flow has generated a significant number of methods existing in the literature. These existing methods can be divided into three types, that is, parametric models, nonparametric models, and hybrid models.

Parametric models are generally based on statistical theory which needs to ensure the time series is in a stationary state. Typical parametric models such as Kalman filtering [2] which is excellent in dealing with traffic flow with noise disturbance. Grey forecasting model [3] can achieve high prediction accuracy even when the number of samples is relatively small. The autoregressive integrated moving average model [4] adopts difference to transform traffic flow into a stationary sequence. Furthermore, seasonal factor is introduced to improve the ability to extract traffic flow periodic features. The proposed seasonal autoregressive integrated moving average [5] is more practical in short-term traffic flow

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prediction. However, parametric models are often unsatisfactory because of the defects in dealing with nonlinear and large-scale traffic flow [6].

Non-parametric models have advantages in dealing with complex traffic flow due to the strong nonlinear fitting ability and self-learning mechanism. Gradient Boosting Decision Tree [7] and Random Forest [8] have been widely used in traffic flow prediction because of the flexibility and portability. Support vector machine [9,10] is effective to solve the regression problem of higher dimensional features because of the strong nonlinear convergence performance. The Artificial Neural Network [11] has gradually become the main approach for traffic flow prediction because of the flexible framework and strong learning ability. Deep learning methods such as RNN [12] and its variant GRU [13] and LSTM [14,15] are widely adopted in short-term traffic flow prediction.

Existing studies [16–18] have demonstrated that hybrid models are more accurate and more applicable than stand-alone models. The defects of a single model can be overcome by the reasonable combination of models, which makes the hybrid model become the most widely used method in short-term flow prediction.

In the process of data collection, traffic flow is often interfered by exogenous factors, such as weather conditions and detector status. Therefore, the collected raw data usually contains a lot of noise which will degrade the performance of models [19]. A decomposition approach for traffic flow can reduce the impact of noise and improve prediction accuracy [20]. Moreover, this approach provides a framework for hybrid modeling.

There are two dominating decomposition methods currently, one is multi-scale decomposition which migrated from the field of signal processing. Hamad et al. [21] combined empirical mode decomposition (EMD) with neural networks to predict short-term travel speed in real-life loop detector data. Tang et al. [22] demonstrated that ensemble empirical mode decomposition (EEMD) is superior to EMD from the aspect of de-noising. Jiang et al. [23] developed a hybrid model by combining EEMD and grey support vector machine which is especially suitable for short-term prediction. Wang et al. [24] proposed a semi-supervised hybrid method based on K-means and wavelet decomposition (WD) which outperformed the traditional model in terms of robustness. Mehmet et al. [25] concluded that a hybrid model with WD has higher accuracy than that with EMD by selecting correct wavelet types. Cheng et al. [26] discovered that an adaptive network-based fuzzy inference system can achieve higher accuracy by introducing wavelet packet decomposition (WPD). Although the multi-scale decomposition is an adaptive approach that can separate the traffic flow without manual intervention, each execution requires integrating the current data with the historical data before decomposition. In the case of dealing with large-scale data, it is relatively inefficient. Moreover, such an approach does not encompass a sound theoretical underpinning for the IMFs.

The other one is based on spectrum analysis which attempts to decompose the traffic flow into several preselected components. Zhang et al. [27] adopted the combination of sine and cosine to model the periodic part, the proposed hybrid model ARIMA-GARCH represented the deterministic part and volatility respectively. Luo et al. [28] utilized discrete Fourier transform to decompose traffic flow and predicted residual part by support vector regression. Chen et al. [29] employed Fourier transform to indicate the periodic part and modeled the volatility by supervised learning. Zhou et al. [30] proposed a novel method to separate traffic flow into periodic and trend parts by locally weighted regression. Zhang et al. [31] adopted accumulated deviation to model trend component. Ghosh et al. [32] treated the trend part as a daily average. Principal component analysis [33,34] is another frequently used definition approach for traffic trend which distinguishes the trend and the deviations. This method overcomes the defect of being unable to offer the physical interpretation for components reasonably. However, the decomposition process should be supervised manually and cannot be performed adaptively.

Considering both the advantages and disadvantages of the foregoing decomposition methods, we can come to a conclusion that a suitable method should appropriately provide a better perspective on the composition of traffic flow, as well as decompose the traffic flow adaptively without manual supervision. Motivated by the foregoing researches, we propose a novel method that combines the advantages of the two methods. we treat the periodic component of traffic flow as a function of time and formed as a Fourier series. We replacing the frequency with instantaneous frequency obtained by Hilbert–Huang Transform (HHT) so as to present the time variants patterns from short-term flow adaptively. Extracting the residual component by EMD and the volatile component is left. Parametric and nonparametric approaches are applied to these components according to their characteristics and the hybrid model is proposed.

The main contributions of this paper are summarized as follows:

- (1) We provide a novel traffic flow decomposition and modeling methodology. TSD adaptively extract time variants patterns and provide a sound theoretical underpinning for the composition of traffic flow from the perspective of time evolution.
- (2) The experimental results show the superiority of the hybrid model in short-term traffic flow prediction in terms of RMSE and MAPE. The supervised learning ensures the single-step prediction performance of the hybrid model while the periodic and residual components can significantly restrain the error growth in multi-step prediction. Furthermore, benefited from decomposition, TSD presents extremely high tolerance for missing data.
- (3) TSD is a universal framework for time series which can be applied to any time series with the periodic feature. It adaptively represents better insights into the intrinsic structures and evolution trend of the time series, such as passenger flow, runoff and power load.

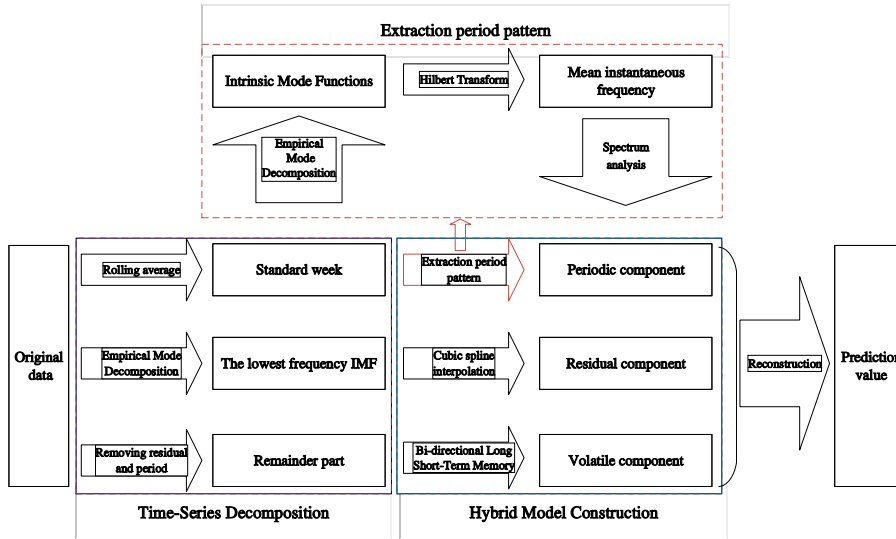


Fig. 1. The framework of TSD.

The rest of the paper is organized as follows. The Section 2 introduces the composition of traffic flow and the methodology of these components. The Section 3 introduces the details of decomposition and presents physical explanations of the periodic patterns. In the Section 4, the proposed hybrid model is compared with baseline models and hybrid models in a real-world dataset from PeMS to validate the effectiveness. The Section 5 gives the conclusions and the focus for further research.

2. Methodology

Conventional traffic flow prediction methods assume the current value is strongly related to the historical value with constant variance. An alternative way is to treat the traffic flow as the superposition of multiple components [35]. In this paper, the structure of the traffic flow is treated as the superposition of the periodic component, residual component and volatile component. The periodic component is seasonal mode which is affected by commuting private cars and public buses; The residual component is an overall evolutionary trend influenced by significant events such as local policy and road rebuilding. The volatile component is a jumble of uncertainties which is determined by external factors, such as bad weather, sensor failure and occasional events. Then the traffic flow can be described as follow.

$$X(t) = P(t) + R(t) + V(t) \quad (1)$$

where $X(t)$ is observed traffic flow during time interval t , $P(t)$ is the periodic component. $R(t)$ is the residual component. $V(t)$ is the volatile component.

Fig. 1 illustrates the overall framework of the proposed hybrid model. The composition of traffic flow and methodology will be described in detail later.

2.1. Periodic component

The periodic component is commonly treated as a trigonometric regression function on the combination of sines and cosines to reproduces the periodic patterns. However, this approach requires the assumption that the periodic component has a uniform frequency. Existing studies [35,36] prove that EMD can extract varying time-variant patterns and corresponding instantaneous frequencies presents physical interpretation of passenger flow. In this paper, we propose a novel periodic modeling approach. The time-variant patterns are transformed into frequencies and described via Fourier series. Compared with conventional periodic models, the proposed approach reduces the parameters to be estimated and the order of regression is rest with the number of IMFs. This means that TSD is computationally cheaper and adaptive. The three stages of periodic component modeling are described as below:

Stage 1: Empirical mode decomposition

In Stage 1, traffic flow is separated into IMFs at different time scales by EMD which represent the local characteristics of traffic flow, and the process of decomposition can be described as follows:

1. Determine all local maximum and minimum points of the original signal $X(t)$, Cubic spline interpolation is employed to fit all the local maximum and minimum points., The upper and lower envelopes formed by cubic spline difference are denoted as $u_0(t)$ and $l_0(t)$ respectively.
2. Calculating the mean value of the upper and lower envelope

$$m_0(t) = \frac{(u_0(t) + l_0(t))}{2} \quad (2)$$

3. Subtract the mean value from original data to obtain the proto-intrinsic mode function $h_1(t)$:

$$h_1(t) = X(t) - m_0(t) \quad (3)$$

4. If the proto-intrinsic mode function satisfies the shifting criteria, $h_1(t)$ is denoted as the i th IMF and substitutes residue $r_1(t)$ for the original time series $X(t)$. Otherwise, the above steps will continue unless $h_1(t)$ becomes an IMF.

$$r_1(t) = X(t) - h_1(t) \quad (4)$$

5. The above step is repeated on every subsequent residual till every residue satisfies one of the termination criteria. First, the $r_i(t)$ is smaller than the predetermined threshold or becomes a monotonic function; Second, the number of zero crossings and extrema is the same as that of the successive sifting step.
6. Finally, the original data $X(t)$ can be decomposed into k modes and a residue as follows:

$$X(t) = \sum_{i=1}^k IMF_i + r_k(t) \quad (5)$$

Stage 2: Hilbert Transform

In Stage 2, Hilbert Transform is employed to obtain the analytical signal and frequency distribution which can be utilized to calculate the instantaneous properties of traffic flow. Eq. (8) demonstrates a strict bijection between time and instantaneous frequency which means a time series can be characterized a unique frequency. The period is the inverse of the corresponding frequency. In this case, we regard the mean instantaneous frequency of the analytic signal as a potential periodic pattern of traffic flow.

1. Performing the Hilbert Transform for i th $IMF_i(t)$ which is defined as the convolution of $IMF_i(t)$ with $1/t$.

$$H[IMF_i(t)] = \frac{1}{\pi} P \int_{-\infty}^{+\infty} \frac{IMF_i(\tau)}{t - \tau} d\tau \quad (6)$$

where P indicates the Cauchy principal value of singular integral. t is the time variable and τ is the time interval.

2. The analytic signal $Z(t)$ can be described as:

$$Z_i(t) = a_i(t) \times e^{i\theta_i(t)} \quad (7)$$

where $a_i(t) = \sqrt{IMF_i(t)^2 + H[IMF_i(t)]^2}$ indicates the instantaneous amplitude and $\theta_i(t) = \arctan(H[IMF_i(t)]/IMF_i(t))$ indicates the instantaneous phase.

3. At last, the mean instantaneous frequency of traffic flow can be defined as:

$$\omega_i = \frac{1}{n} \sum_{i=1}^n \frac{d\theta_i(t)}{dt} \quad (8)$$

Stage 3: Spectrum analysis

In Stage 3, spectrum analysis is adopted to model the periodic component which is treated as a combination of periodic sines and cosines. The periodic part is generally regarded as a periodic function and can therefore be represented by a Fourier series. Different from the conventional method, the fitting order k is determined by the number of modes obtained in stage 1, and the mean instantaneous frequency ω_i obtained in stage 2 replaces the frequency which avoids relying on experience to set parameters. The periodic component can be modeled as:

$$P(t) = c + \sum_{i=1}^k a_i \cos(2\pi \omega_i t) + b_i \sin(2\pi \omega_i t) \quad (9)$$

where k is the index of individual periodic pattern, c is the phase of periodic component, a_i , b_i are the amplitude of i th individual periodic pattern, ω_i is the frequency obtained by HHT.

The pseudo-code of the periodic component is given in Algorithm 1.

Algorithm 1 Periodic component

```

Input: original data vector:  $\mathbf{x}$ 
Output: Periodic component:  $P(t)$ 
Initialization:  $IMF \leftarrow 0, i \leftarrow 0, r_i \leftarrow 0, h_i \leftarrow 0$ 
Function EMD(  $\mathbf{x}$  )
while  $r_i$  not satisfy  $criteria_1$  do
    While  $h_i$  not satisfy  $criteria_2$  do
         $[IndMin, LocalMin] \leftarrow \min(\mathbf{x})$ 
         $[IndMax, LocalMax] \leftarrow \max(\mathbf{x})$ 
         $l_i \leftarrow spline(IndMin, LocalMin)$ 
         $u_i \leftarrow spline(IndMax, LocalMax)$ 
         $m_i \leftarrow (l_i + u_i) / 2$ 
         $h_{i+1} \leftarrow \mathbf{x} - m_i$ 
    end while
     $IMF_{i+1} \leftarrow h_{i+1}$ 
     $r_{i+1} \leftarrow \mathbf{x} - IMF_{i+1}$ 
     $i \leftarrow i + 1$ 
end while
Return  $IMF_i, r_i$ 
Function HT(  $IMF_i$  )
 $H[IMF_i(t)] = \frac{1}{\pi} P \int_{-\infty}^{+\infty} \frac{IMF_i(\tau)}{t - \tau} d\tau$ 
 $\theta_i(t) = \arctan(H[IMF_i(t)] / IMF_i(t))$ 
 $\omega_i \leftarrow \frac{1}{n} \sum_{i=1}^n \frac{d\theta_i(t)}{dt}$ 
Return  $\omega_i$ 
 $P(t) \leftarrow c + \sum_{j=1}^k (a_j \cos(2\pi\omega_j t) + b_j \sin(2\pi\omega_j t))$ 
Return  $P(t)$ 

```

2.2. Residual component

The residual component, which is called the trend component as well, is the deterministic part of the traffic flow which is not affected by accidents or changes in commuting patterns occasionally. The residual component is the “solid part” that will only be transformed by some profound events, such as policy changes and road rebuilding. Existing studies treat it in some simple methods like simple average trend and cumulative difference. However, the residual component is a representation of overall change and should be described at a wider time scale. Studies [37] indicate that the IMF with the lowest frequency represents the trend of the whole time series. So the residual component can be defined as $R(t) = r_k(t)$. The residual component generated by EMD is a stationary curve that can be modeled by linear fitting.

2.3. Volatile component

The volatile component of traffic flow is influenced by variable exogenous factors. Some of them like severe weather and public events occurs with some foreboding in recent historical data, while others like car incident and sensor failure occur completely randomly and unpredictably. In this case, we assume the volatile component as a remainder obtained by separating periodic component and residual component from origin data, that is $V(t) = X(t) - P(t) - R(t)$.

These complicated situations make the conventional model powerless, the prediction of the volatile component is still a challenging issue so far. Chen et al. [29] assert that the volatile component should be modeled by a non-parametric method and LSTM is employed. Li et al. [38] found that Bi-directional Long Short-Term Memory (BiLSTM) could better capture temporal correlation compared with LSTM because of the mechanism of bi-directional training simultaneously.

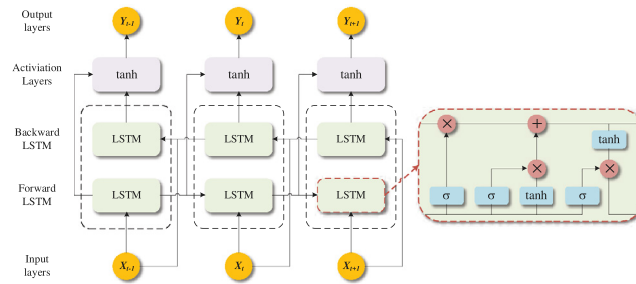


Fig. 2. BiLSTM neural network structure.

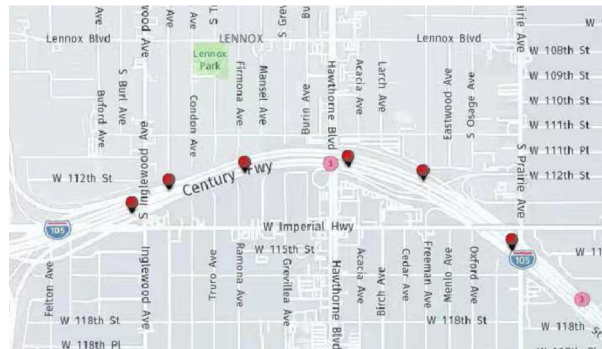


Fig. 3. The locations of detectors.

Related researches [9,38,39] demonstrate that BiLSTM has advantages over other supervised learning methods as well (i.e. RF, SVM, GRU). In addition, the bi-directional training mechanism plays a role in data enhancement which is more beneficial for the prediction of small-scale datasets in this paper. These special advantages drive us to adopt BiLSTM to predict the volatile component. The process and structure can be expressed as Eq. (10) and Fig. 2.

$$\begin{aligned}
\mathbf{i}_t &= \sigma(\mathbf{W}_{xti}\mathbf{X}_t + \mathbf{W}_{hti}\mathbf{H}_{t-1} + \mathbf{W}_{cti}\mathbf{C}_{t-1} + \mathbf{b}_i) \\
\mathbf{f}_t &= \sigma(\mathbf{W}_{xtf}\mathbf{X}_t + \mathbf{W}_{htf}\mathbf{H}_{t-1} + \mathbf{W}_{ctf}\mathbf{C}_{t-1} + \mathbf{b}_f) \\
\mathbf{c}_t &= \mathbf{f}_t\mathbf{c}_{t-1} + \mathbf{i}_t \tanh(\mathbf{W}_{xtc}\mathbf{X}_t + \mathbf{W}_{htc}\mathbf{H}_{t-1} + \mathbf{b}_c) \\
\mathbf{o}_t &= \sigma(\mathbf{W}_{xto}\mathbf{X}_t + \mathbf{W}_{hto}\mathbf{H}_{t-1} + \mathbf{W}_{cto}\mathbf{C}_{t-1} + \mathbf{b}_o) \\
\mathbf{H}_t &= \mathbf{o}_t \tanh(\mathbf{C}_t)
\end{aligned} \tag{10}$$

where $\mathbf{i}_t, \mathbf{f}_t, \mathbf{c}_t, \mathbf{o}_t$ are the output of input gates i , forget gates f , input gates o and cell state c at time t respectively; $\mathbf{W}_{x_i}, \mathbf{W}_{x_{if}}, \mathbf{W}_{x_{io}}, \mathbf{W}_{x_{ic}}$ are the bias of input $\mathbf{X}_t$ and input gates i , forget gates f , input gates o and cell state c respectively; $\mathbf{W}_{h_i}, \mathbf{W}_{h_{if}}, \mathbf{W}_{h_{io}}, \mathbf{W}_{h_{ic}}$ are the bias of hidden layer output $\mathbf{H}_t$ and corresponding gated units; $\mathbf{W}_{c_i}, \mathbf{W}_{c_{if}}, \mathbf{W}_{c_{io}}$ are the bias of cell state output $\mathbf{C}_t$ and corresponding gated units; $\mathbf{b}_i, \mathbf{b}_f, \mathbf{b}_o, \mathbf{b}_c$ is the bias vector; σ is the activation function.

3. Experiment

3.1. Preliminaries

Data descriptions and preprocessing: To provide a fair comparison, we conduct experiments on an open-access dataset of Caltrans Performance Measurement System (PeMS). The objective of the PeMS project is to collect real-time freeway data from freeways in California. The IDs of these detectors are 767 838, 760 112, 760 074, 760 080, 716 414 and 760 101 (for convenience, we named them Sections 1 to 6 respectively). The locations of detectors are consecutive and displayed in Fig. 3. The sampling period ranges from May 1, 2017 to June 30, 2017. The traffic flow data are aggregated 5-min historical data. To get a general model, we will not exclude weekends, so each location contains 2016 time slots per week. The missing data (about 1%) have been imputed by the moving average method. At last, we obtain a traffic flow matrix $\mathbf{X}^{17568 \times 6}$ and then the former 80% is chosen as training set $\mathbf{X}_{train}^{14055 \times 6}$, the remains is chosen as test set $\mathbf{X}_{test}^{3513 \times 6}$. The subsequent experiment results are based on the test set.

Prediction performance evaluation: Three metrics are adopted to evaluate the prediction performance comprehensively. Root mean square error (RMSE) reflects the absolute difference between the predicted value and observed value and be sensitive to the extremely large or extremely small error, but it cannot provide a fair comparison in different situations.

Table 1
Mean instantaneous frequency and transformed period.

	IMF ₁	IMF ₂	IMF ₃	IMF ₄	IMF ₅	IMF ₆	IMF ₇
ω_i (times/5-min)	0.2191	0.0966	0.0251	0.0103	0.0035	0.0011	0.0004
T_i (h/cycle)	0.3803	0.8627	3.3203	8.1005	23.9557	78.4430	181.2444

Mean absolute percentage error (MAPE) offers a relative deviation between the predicted value and the observed value by measuring percentage error which can effectively make up for the deficiency. The determination coefficient (R^2) reflects the degree of similarity between the predicted value and observed value, these metrics are calculated by the following formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n |x_t - \hat{x}_t|} \quad (11)$$

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{x_t - \hat{x}_t}{x_t} \right| \quad (12)$$

$$R^2 = 1 - \frac{\sum_{t=1}^n (x_t - \hat{x}_t)^2}{\sum_{t=1}^n (x_t - \bar{x}_t)^2} \quad (13)$$

where x_t is the observed value, $\hat{x}_t$ is the predicted value and $\bar{x}_t$ is the average value of observed value.

Baseline Model: To verify the effectiveness of the TSD and corresponding hybrid model, commonly used prediction models are introduced for experiment and comparison, that is Long short-term memory (LSTM) whose parameters are consistent with BiLSTM, Random Forest (RF), and least squares support vector machine (LSSVM). RF adopts an empirical formula $N = \sqrt{S - 1}$ (where S indicate the size of samples) to determine the number of subtrees N , while LSSVM adopts RBF kernel and the hyperparameters are optimized by particle swarm optimization (PSO).

3.2. Composition and analysis

Periodic component: It is impractical to regress the period component over the entire timeline [29], especially in such a short sampling interval. In this paper, the periodic component is treated as the combination of sines and cosines, in which the periodic pattern is no more than one week generally. In this case, we propose an assumption of “standard week”, that is an average value collected on the same site at the same daily time interval but from neighboring weeks (rolling average) which can smooth the variation caused by short sampling intervals. Then, EMD is employed to the standard week and we obtain 8 *IMFs*. It is worth noting that the residue is excluded and only 7 components are applied to the experiment and displayed in Fig. 4. As mentioned above, each IMF has the well-behaved Hilbert transform property and represents a certain periodic pattern of traffic flow. However, we cannot detect these patterns in the form of IMF. IMF (a time series) can be converted to instantaneous frequencies (a numerical value) by Hilbert transform. The mean instantaneous frequency can present the physical interpretation of patterns intuitively.

It is obscured to detect the physical meaning of each periodic pattern because of the sampling interval of 5-min. For an intuitive observation, mean instantaneous frequency is transformed into period T in hours (i.e. $T_1 = 0.3803$ indicates 0.3803 h per cycle). The transformation formula can be described as $T = 5 \times 60^{-1} \times \omega^{-1}$. Table 1 lists the mean instantaneous frequency extracted from IMFs and the corresponding transformed period. T_1 and T_2 are 0.3803h (about 30 min) and 0.8627 h (about 60 min) per cycle respectively which represents the periodic pattern of the top peak period (rush hour). T_3 is 3.3203 h per cycle indicate an ordinary peak pattern. Due to the traffic flow fall dramatically from 10:00 PM to 6:00 AM the next day, the effective service hour on the expressway can be considered as 16 h. Then T_4 which is 8.1005 h per cycle, the half of service hour, could be regarded as the semi-service pattern of traffic flow. T_5 represents the daily pattern of traffic flow distinctly because of its period of 23.9557 h per cycle. It is worth noting that the frequency of daily patterns has a significant effect on regression. At last, T_6 and T_7 could be defined as the long-term pattern of traffic flow. The period of these patterns are 78.4430 h (about semi-week pattern) and 181.2444 h (week pattern) per cycle, respectively.

Fig. 5 represents the detail of the periodic component. After excluding the residue from the standard week, we can notice that the remaining traffic flow is shifted to the vicinity of 0 from Fig. 5(b). This means that we have succeeded in removing the trend of the traffic flow. There is still residual noise perturbation in the standard week. To avoid overfitting noise, wavelet de-noising is performed before regression. We adopt sym4 wavelet to perform 8-level de-noising, which is excellent in processing the periodic signal. The de-noising method is Bayes, and the rule of threshold is set as the median. From the observation of Fig. 5(c), wavelet de-noising smooths the fluctuation of time series and weakens the influence of noise on period modeling.

Then the periodic component is regressed by Eq. (9). The parameter of k is determined by the number of IMF, ω is replaced by mean instantaneous frequency correspondingly. The estimated parameters of period cosines and sines are

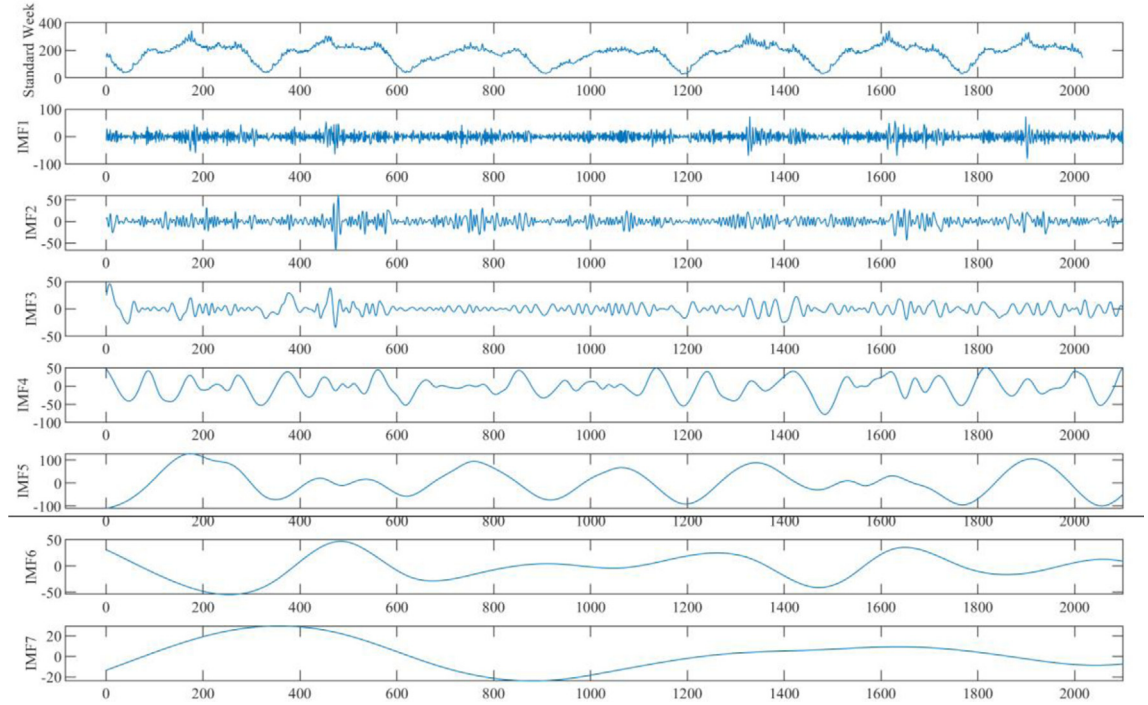


Fig. 4. The standard week and corresponding IMFs.

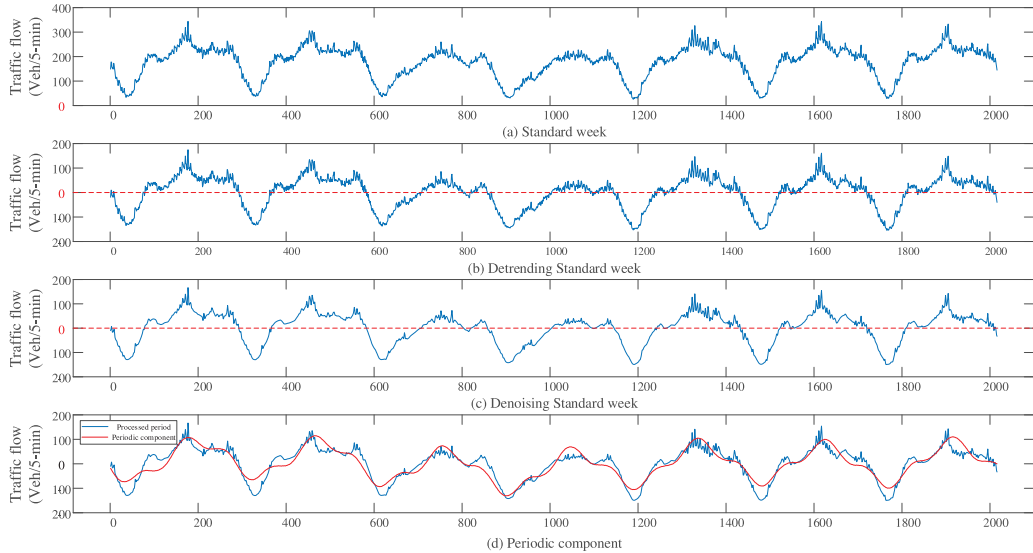


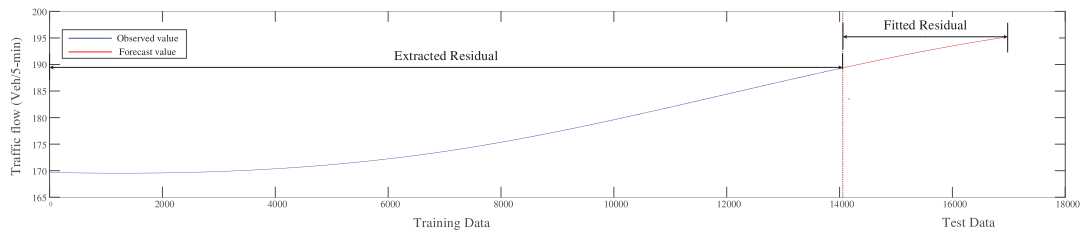
Fig. 5. The modeling process of the periodic component. (a) Standard week (b) Standard week after excluding trend (c) Standard week after denoise (d) Periodic component compared with processed standard week.

presented in Table 2 with corresponding frequency. Fig. 5(d) compares the periodic component in which R^2 reaching 82.64% with standard week after processing. We should point out that excessively high R^2 may imply a fact that the noise is over fitted. In this case, R^2 should be used as an auxiliary evaluation approach. At last, the periodic component is periodically extended over the entire dataset to represent the periodic part of traffic flow.

Residual component: As mentioned above, the residual component should be described in the widest time scale. So EMD is performed in the whole dataset, rather than in the standard week. Then the residual component is the lowest frequency component obtained by EMD. Fig. 6 shows the extracted residual component in training data. Then cubic spline

Table 2
Estimated parameters.

	IMF_1	IMF_2	IMF_3	IMF_4	IMF_5	IMF_6	IMF_7
ω_i	0.2191	0.0966	0.0251	0.0103	0.0035	0.0011	0.0004
a_i	-29.07	1.48	-0.94	-0.01	0.18	-14.66	22.29
b_i	-68.91	-25.69	-1.09	-0.07	-0.18	0.82	-11.76
$c = 6.985$							

**Fig. 6.** Residual component.**Table 3**
Comparison results in each section.

Model		Section 1	Section 2	Section 3	Section 4	Section 5	Section 6
BiLSTM	RMSE	18.3185	23.1903	26.8143	27.8902	28.6899	21.6161
	MAPE	0.1116	0.0558	0.0836	0.0999	0.0673	0.0617
	R^2	0.9291	0.9707	0.9514	0.9180	0.9460	0.9673
TSD-BiLSTM	RMSE	16.9330	21.5177	25.0949	26.0101	26.6537	19.8541
	MAPE	0.1019	0.0530	0.0778	0.0943	0.0632	0.0583
	R^2	0.9394	0.9746	0.9579	0.9262	0.9528	0.9722

interpolation is adapted to fit the residual component and generates the forecast value of test data. It can be observed that in the absence of major events, the traffic flow in this section will increase approximately linearly in the future.

Volatile component: Different from the periodic and residual component, intense variability of the volatile component makes the conventional parameter method powerless for practical prediction applications. As discussed above, we seek help from BiLSTM and prediction steps are presented as follows

Step 1. Preparing training data: The datasets are split into 80% training set and 20% test set, then the data is normalized to $[0,1]$, which can accelerate the convergence of BiLSTM.

Step 2. Training parameters: The number of epochs is set as 50, and the initial learning rate is set as 0.005. To prevent the learning rate from being too large, the learning rate is dynamically adjusted by decreasing 50% for every 20 epochs. The dropout is set as 0.3 to avoid overfitting. We adopt the Adam optimizer with the loss function of RMSE. As for the structure of BiLSTM, we adopt a single hidden layer structure with 32 neurons because of the limited data. The activation function defaults to tanh. The time step is set as 12 and the outputs of multi-step prediction are obtained by direct strategy.

Step 3. Prediction and reconstruction: Finally, we combine the fitted periodic component $\hat{P}_t$ and residual component $\hat{R}_t$ with the predicted volatile component $\hat{V}_t$. The reconstructed traffic $\hat{X}_t$ is denoted as $\hat{X}_t = \hat{P}_t + \hat{R}_t + \hat{V}_t$.

4. Results and analysis

4.1. Prediction performance analysis

To test the improvement of the proposed decomposition method on BiLSTM, TSD is applied to the six sections respectively and compared with the BiLSTM without TSD. The prediction performance is comprehensively evaluated by RMSE, MAPE, R^2 and shown in Table 3. Generally, we notice that the hybrid model outperforms the stand-alone models in all sections. The performance improved by 6.8% to 8.9% in terms of RMSE and 5.9% to 9.5% in terms of MAPE. It is worth noting that the periods of daily pattern reach 26.4519 h and 30.5960 h per cycle respectively in section 4 and section 6. This is regarded as a mode mixing occurred during EMD. Huang [40] believes that mode mixing is often induced by data noise. Although the IMFs of traffic flow cannot be extracted well, the prediction performance of hybrid models is still better than that of the stand-one model in single-step prediction.

Multi-step prediction of traffic flow on several time horizons allows for a wider range of applications in intelligent transportation systems [41]. To validate the effect of TSD on the multi-step prediction, we tested the variation of prediction accuracy for five horizons of 5 min, 15 min, 30 min, 45 min and 60 min respectively. The RMSE and MAPE are given in Fig. 7. It can be found that the longer horizon did not deteriorate the prediction accuracy significantly by introducing

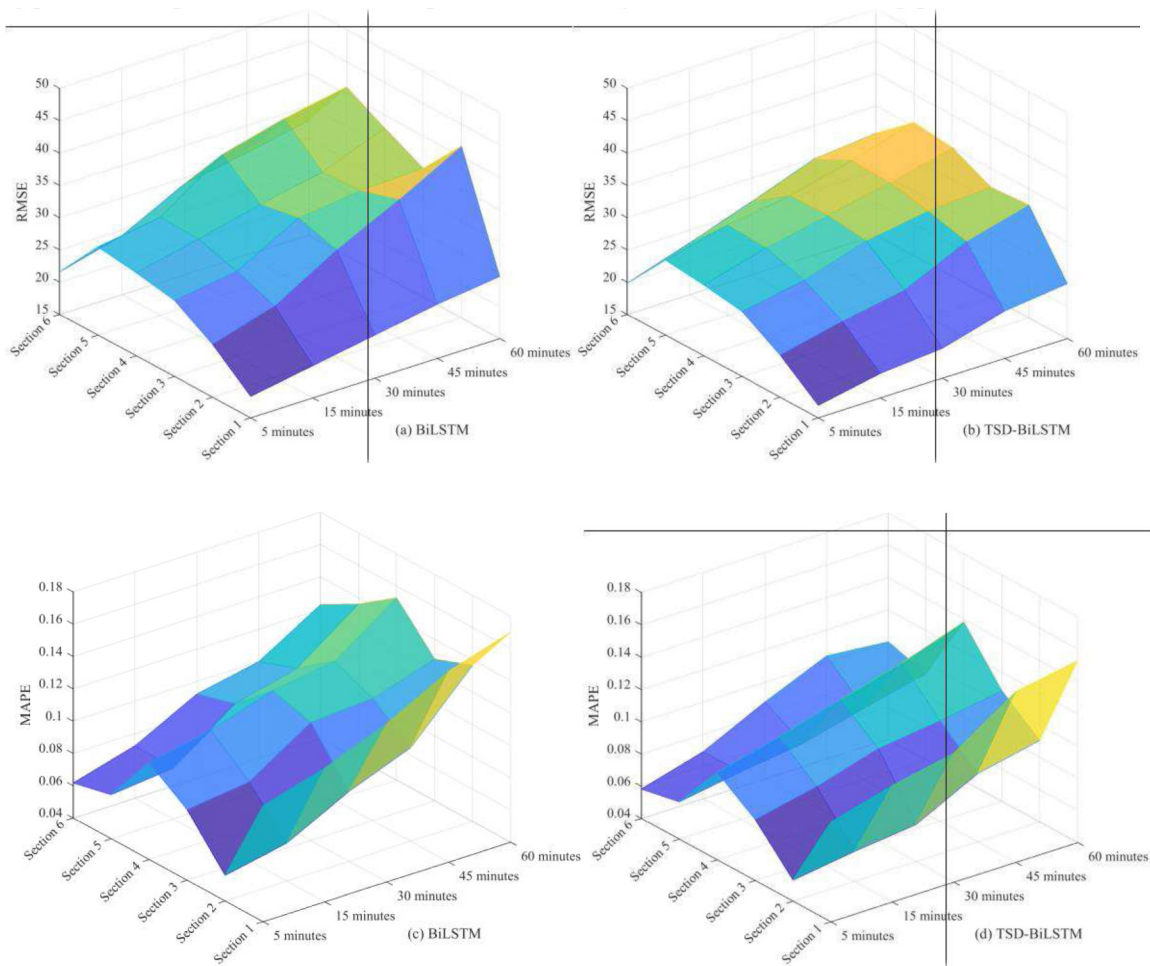


Fig. 7. The results of multi-step prediction. (a) the RMSE of BiLSTM (b) the RMSE of TSD-BiLSTM (c) the MAPE of BiLSTM (d) the MAPE of TSD-BiLSTM.

TSD. On most occasions, the hybrid model performs more stable and accurate than the stand-alone model in multi-step prediction. When the time horizon is more than 30 min (6-step ahead), the effect of TSD on curbing the growth of error becomes more obvious, especially in Section 2 and Section 5, while Section 4 and 6 show a marginal difference. As mentioned above, the periodic component of section 4 and section 6 are not well extracted. This demonstrates that an outstanding periodic component can well make up for the deficiency of BiLSTM in multi-step prediction.

Table 4 summarizes the average results of baseline models and the hybrid model of all sections. These models aim at capturing temporal correlation and spatial correlation is not be considered. Generally, we conclude TSD improves marginal performance for some baseline models (i.e. RF), but at least it will not get worse. The improvement of TSD on these models is not as good as that of BiLSTM, which may be because the traditional model is not competent to predict the volatile component in a limited dataset. Of all the baseline models, LSTM performed the worst, while RF is similar to LSSVM. Due to limited data, BiLSTM has better performance because of the mechanism of bi-directional training. The hybrid models present the similar results as the baseline model which means that a preeminent baseline model is the foundation for single-step prediction.

The fusion of spatial-temporal correlation would further improve forecasting performance, and we have carried out extensive tentative experiments aimed at introducing the spatial correlations of road segments into our hybrid model. We select a classical model CNN-LSTM and two state-of-the-art models Conv-BiLSTM [38] and T-GCN [42] to integrate with TSD. The hyperparameters of two state-of-the-art models are consistent with corresponding literatures, CNN-LSTM adopts the same parameters as Conv-BiLSTM considering the similar structure. The results are listed in Table 5 and we found an apparent performance degradation after integrating with TSD. We speculate that this performance degradation is due to our underlying assumptions about traffic flow that are inconsistent with the ones they adopt. The foregoing methods regard the traffic flow as an indivisible whole and a convolution approach is employed to capture the spatial correlation. However, the traffic flow is regarded as the superposition of multiple components in our research, and only complicated component like volatile component is recommended to predicted by machine learning. The volatile component is generally

Table 4
Comparison of temporal models and hybrid models.

Temporal correlation model		RMSE	MAPE	R ²
Baseline model	LSTM	32.1963	0.1159	0.8937
	RF	30.3735	0.0998	0.9075
	LSSVM	29.9575	0.0967	0.9106
	BiLSTM	24.4199	0.0800	0.9471
Hybrid model	TSD-LSTM	31.7088	0.1106	0.8975
	TSD-RF	30.3207	0.1041	0.9079
	TSD-LSSVM	29.6626	0.0958	0.9127
	TSD-BiLSTM	22.6772	0.0747	0.9538

Table 5
Comparison of spatial-temporal models and hybrid models.

Spatial-temporal correlation model		RMSE	MAPE	R ²
Baseline model	CNN-LSTM	33.1955	0.1199	0.8958
	Conv-BiLSTM	24.2581	0.0742	0.9488
	T-GCN	26.6857	0.0879	0.9275
Hybrid model	TSD-CNN-LSTM	43.1481	0.1513	0.8461
	TSD-Conv-BiLSTM	30.2314	0.1029	0.8973
	TSD-T-GCN	33.3572	0.1199	0.8958

Table 6
Improvement of TSD in RMSE.

Models	Increment of error		Improvement
	No TSD	With TSD	
LSTM	15.6293	8.4369	85.25%
RF	17.3899	9.3605	85.78%
LSSVM	17.2543	9.2444	86.65%
BiLSTM	11.0939	8.9130	24.47%

not influenced by the volatile component of the adjacent segments, namely a convolution or graph convolution is not suitable for our framework. We are still looking for an entry point to introduce spatial-temporal correlation in further research.

We employ baseline models and hybrid models respectively for multi-step prediction in all sections as well, and the average results are shown in Fig. 8. The results illustrate that the introduction of TSD can effectively restrain the decline of prediction accuracy. It indicates that the periodic and residual components of TSD ensure stability in multi-step prediction. From the observation of Fig. 8(a), we can found that TSD can not only restrain the rapid increase of error, but also make the prediction result more stable. Compared to other baseline models, BiLSTM performs well at any level, which is why we chose it to deal with the volatile component.

Table 6 lists the increment of error from 5 min (1-step ahead) to 60 min (12-step ahead). With the combination of TSD, the increase in error is controlled within 10 in terms of RMSE. For models with mediocre performance, the improvement is more pronounced. The improvement can be up to 80% in baseline models which demonstrates the significant effect of TSD in improving the multi-step prediction.

4.2. Time cost and robustness analysis

Time cost plays an important role in evaluating the practical performance of models. Compared with conventional models, the introduction of TSD increases the complexity of the model which will inevitably lead to extra time costs. To explore how much TSD affects the time cost of the model, we extract the required running time of each component. It is worth noting that the time costs of non-major processes (i.e. the process of preparing the data) are incorporated into the volatile component for comparison purposes.

The results of time analysis have shown that the extra time cost is fixed about 2.5 s for a single model in Fig. 9(a). Fig. 9(b) presents the time cost distribution after adopting TSD. Machine learning-based predictive models are the major contributor to time cost, accounting for 64%–98%. Then is the model of the periodic component, because spectrum analysis and de-noising process are time-consuming which accounts for 2%–6% of the time cost. The time cost of other parts (i.e. Residual component and decomposition process) is less than 1%. Overall, the introduction of TSD does not add significantly to the time cost which is acceptable relative to the significant performance improvement.

There are inevitably data missing during the data collection process due to the malfunction of detectors. To test the robustness of TSD, we simulate the data missing rates for five horizons. We refer to the study of Chen et al. [43] and adopt the mode of missing completely at random (MCAR) to simulate the missing data situation.

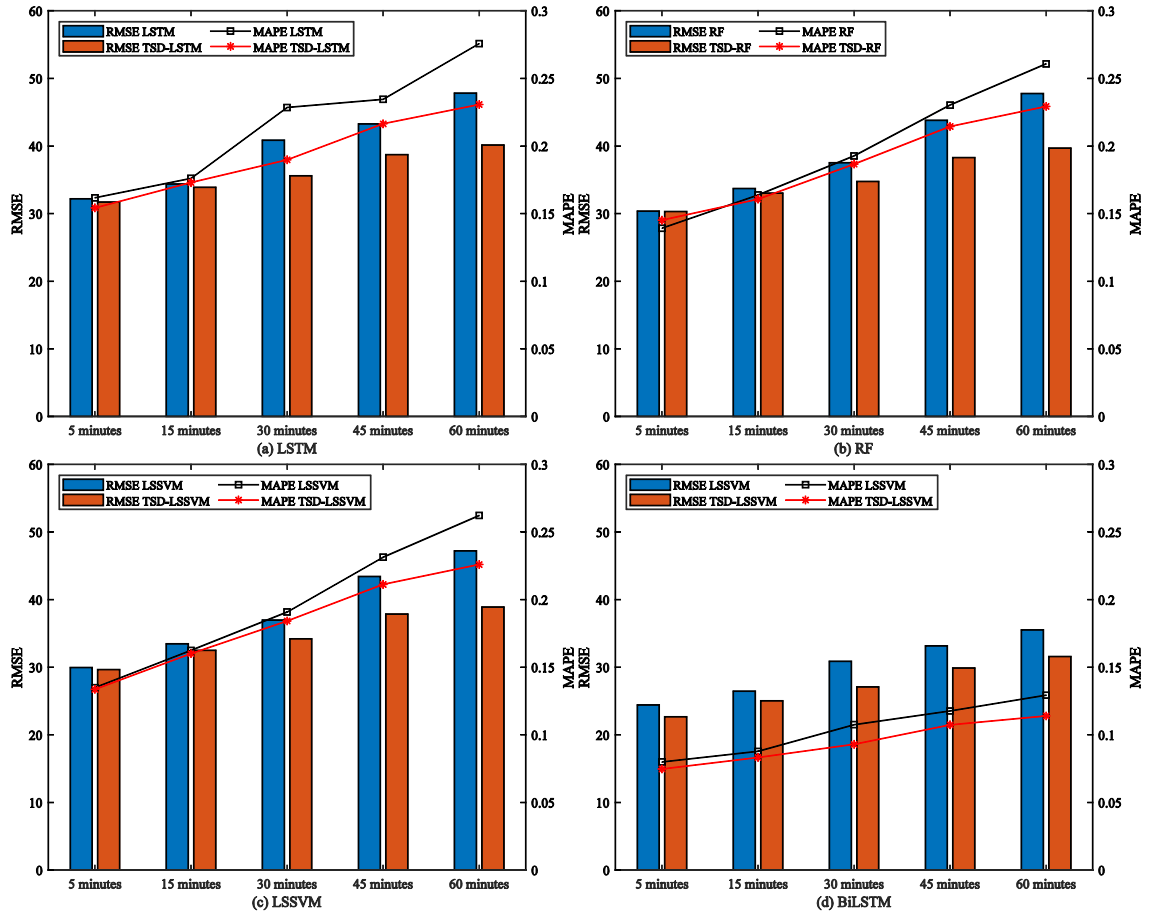


Fig. 8. Prediction results of different horizons.

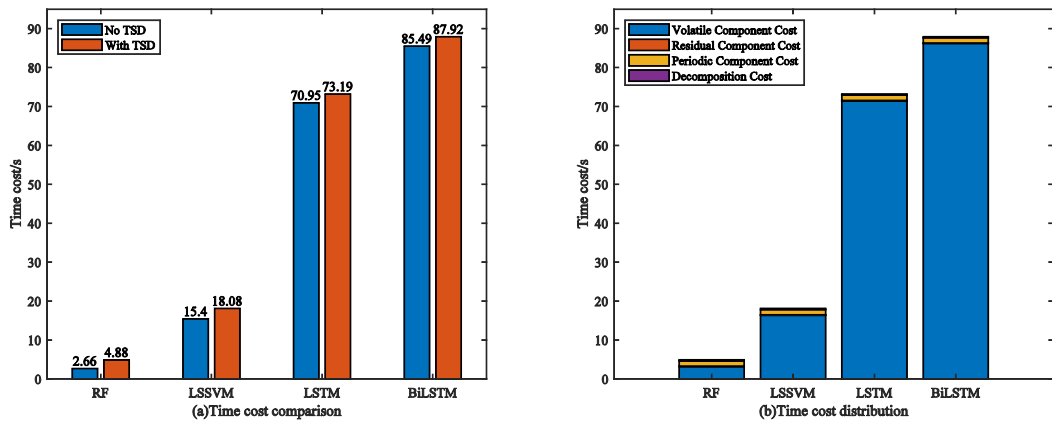


Fig. 9. Time cost for each model. (a) Time cost comparison (b) Time cost distribution.

The results of robustness analysis are shown in Fig. 10 which presents an inspiring conclusion. The introduction of TSD enables models an extremely high tolerance for outliers. As the missing rate grows, the error maintains a relatively small level after adopting TSD, while the error will grow rapidly without adopting TSD. This is because TSD separates the traffic flow into three components, thus avoiding the impact of missing data. For periodic and residual components, we model them via stable parametric models which are hardly affected by occasional missing data. For the volatile component, we adopt a machine learning model to extract the uncertainties. Indeed, the missing data are originally regarded as part of

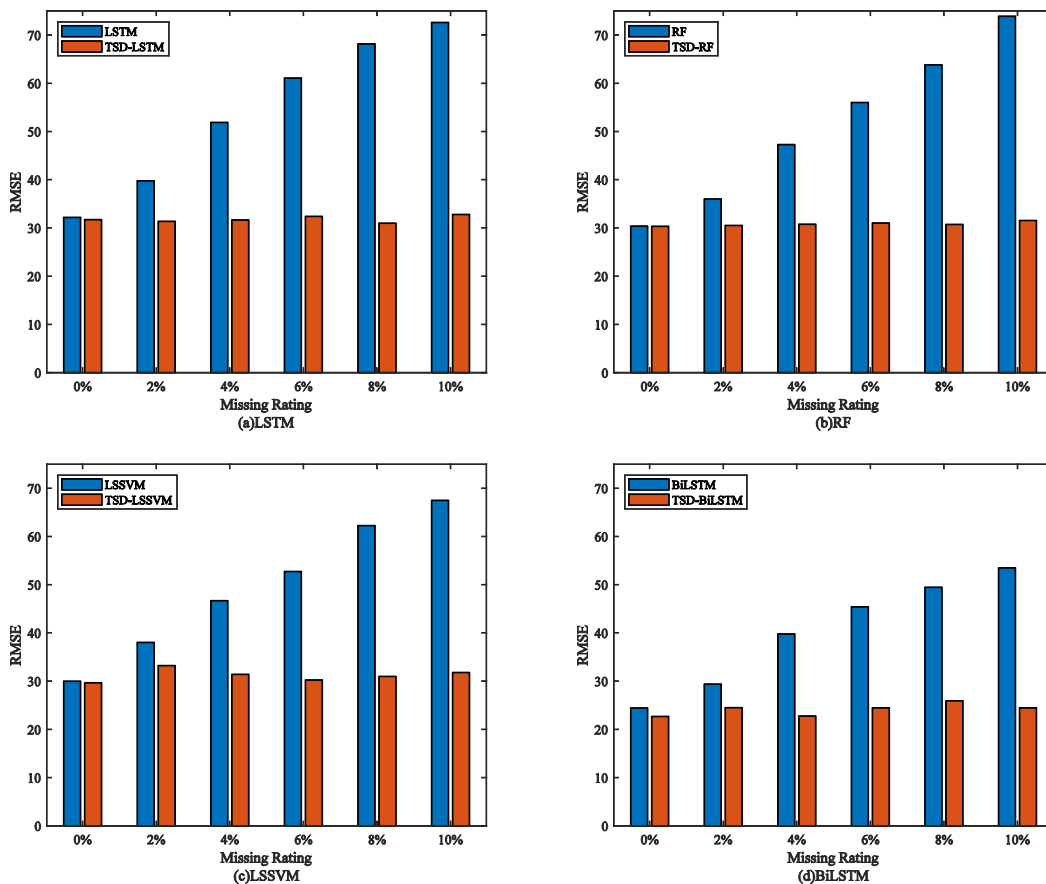


Fig. 10. Robustness analysis (a) LSTM (b) RF (c) LSSVM (d) BiLSTM.

the volatile component. That is, the outliers are abnormal inputs for the conventional model, while the outliers are normal inputs for our model.

5. Conclusion

This paper proposes a novel decomposition and modeling framework named time-series decomposition (TSD) for short-term traffic flow which combines HHT and supervised learning. TSD can extract the periodic pattern of traffic flow and provide physical explanations for each pattern. Based on periodic patterns, we offer a novel perspective to represent better insights into the intrinsic composition of the traffic flow. TSD adaptively decomposes traffic flow into periodic component, residual component and volatile component, then models are constructed according to component feature. The proposed hybrid model is evaluated on a real-world traffic flow dataset comparing with baseline models. The result indicated that the volatile component modeled by BiLSTM guarantees the accuracy of single-step prediction while periodic and residual components generate better prediction reliability in terms of multi-step prediction. Furthermore, TSD presents an extremely high tolerance for outliers, which means that our approach is applicable to a wider range of real-world situations.

We should point out that the decomposition process compromises the spatial dependence of adjacent road segments. Namely, a convolution or graph convolution will result in performance degradation of the model. However, extracting spatial dependence is an important method to improve prediction accuracy. We will focus on exploring an entry point to introduce spatial relationships in future research.

CRedit authorship contribution statement

Haichao Huang: Conceptualization, Methodology, Software, Validation, Formal analysis, Writing – original draft. **Jingya Chen:** Resources, Supervision, Funding acquisition. **Rui Sun:** Investigation, Data curation, Writing – review & editing. **Shuang Wang:** Visualization, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported in part by the National Natural Science Foundation of China under Grant 52078190, in part by the Humanity and Social Science Youth Foundation of Ministry of Education of China under Grant 18YJAZH119.

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